

NTU AI 2024–HW3 Report

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Details can be found: [GitHub Repository](#)

Disclaimer: I discussed this homework with *Hao Wei Yei (R13922173)*, so some similarities may exist.

Agents Setup

- For this homework, I modified the original setup by integrating the *Review Analysis Agent* and the *Scoring Agent*. I also utilized regular expressions (regex) to extract all individual scores returned from *Review Analysis and Scoring Agent*, and compute the overall score.
- This optimized setup improves upon the initial design by reducing the likelihood of hallucinations from the Scoring Agent, as it no longer needs to reference the full context when assigning scores.

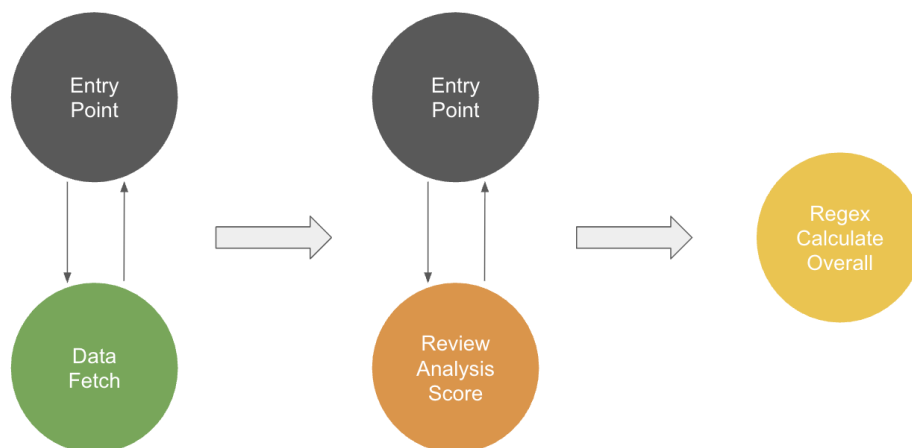


Figure 1: Optimized Agents Workflow

Prompt Design

- I discussed the prompt with *Hao Wei Yei* before passing all public test cases; our prompts may therefore be similar.
- We originally used the three-agent setup with detailed, step-by-step instructions. Although this method retrieved and scored in a expected way, it failed to show its robustness in multiple rounds.
- Runtime logs showed that during review analysis the agent sometimes fetched incomplete reviews, causing scoring errors.
- To fix this, we dropped the step-by-step output—likely overloading the attention mechanism—and now score each review immediately after retrieval, so each score relies only on the most recent sentence.



Figure 2: Original three agents workflow



Figure 3: Our two agents workflow

Case Analysis: Success and Failure

Success Cases

- Example 1: Description and analysis
- Example 2: Description and analysis

Failure Cases

Discuss potential causes of failures.

- Example 1: Description and error analysis
- Example 2: Description and error analysis

Conclusion and Key Modifications

Conclude with the main takeaways and describe any changes you plan to implement to improve performance.